



Phase 1: Conception

Project: Build a Data Mart in SQL

BSC in Computer Science

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Topic: ERM and Data Model for Airbnb-style Accommodation Platform

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1. Introduction

The development of the data-driven solution to a digital accommodation platform like Airbnb requires a well-designed and solid database framework. The Entity Relationship Model (ERM) is at the centre of this framework and makes sure that different data points, such as listings and bookings, user reviews, and administrative functions, are logically and efficiently interconnected. This project aims at modelling a system that would support complex interactions and be scalable, data and usable. Using the experience of success and problems of the own platform architecture of Airbnb, this ERM incorporates entity interaction in the form of reservations, payments, multilingual communication, and complaint management.

The scholarly literature highlights the disruptive nature of Airbnb that changed the conventional hospitality paradigm by offering an informal, peer-to-peer service (Guttentag, 2015). The proper ER model makes the system more transparent and manageable, which can help to avoid the risk of failure to enforce it or bypassing the regulatory framework (Leshinsky & Schatz, 2018). The data-driven approach is applied in this design to support user preferences, property regulations, multilingual access, and booking management, therefore, making the platform competitive and operationally viable.

2. Roles

The ERM has a number of user and system roles that have their own responsibilities in the platform. These are **Guests, Hosts, Administrators, Support Staff, and Employees**. The different roles will be assigned to different entities and relationships to encourage workflows specific to roles.

- End-users are **guests** who search and book listings and leave the reviews. They deal mostly with booking, payment, and review information.
- **Hosts** are people who offer accommodation, handle listing, price, availability, and communication with guests.
- **Administrators** deal with user management, platform configuration and conflict resolution.
- The **Support Staff** assists users in troubleshooting and handling requests and provides first-line support.
- **Employees** are the organizational personnel that keep the platforms running, including the hiring process and internal HR.

These functions are consistent with the results of Gurran (2018), who observed that the process of platform governance and user management is becoming increasingly complicated in peer-to-peer hospitality settings. The addition of support functions also indicates the move to hybrid service models with inbuilt professional support at Airbnb (Cheng & Mackenzie, 2022)..

3. Actions

Each role in the system executes defined actions that shape the interaction flow and functional coverage of the platform:

Guest Actions:

- Browse listings using filter parameters (e.g., price, amenities, availability).
- Make reservations and manage check-in/check-out dates.
- Review past stays, providing qualitative and quantitative feedback.
- Update profile preferences and manage payment history.

Host Actions:

- Create and manage listings by uploading photos, setting prices, and defining house rules.
- Confirm or decline booking requests based on calendar availability.
- Monitor income via autogenerated reports.

Administrator Actions:

- Manage user registration and access credentials.
- Review platform activity logs and configure system-wide settings.
- Manage formal complaints and resolve user conflicts.

Support Staff Actions:

- Address user queries via integrated notification systems.
- Support payment or booking error resolution.
- Escalate unresolved complaints to administrators.

Each of these actions is mapped in the ERM with appropriate foreign keys and relationship cardinalities, ensuring a relationally sound database. As Guttentag (2015) and Prayag et al. (2022) emphasize, clear workflows are essential for maintaining platform trust and regulatory compliance in sharing economy models.

4. Data and Functions

The ERM diagram of the Airbnb-style platform has a properly organized data architecture that supports a wide variety of interactions between users and the system-level operations. The data model focuses on the most important entities: User, Accommodation, Booking, Payment, Review,

and Place, and each of them is related to the particular functional requirements that are necessary to have a dynamic hospitality platform. These functions indicate the requirements of real-world application, including support of multiple languages, secure booking and payment processes, and administrative control.

As a guest, the platform allows a data-informed search and reservation of the listings. The Place and Accommodation objects contain the features of Pricing, Availability, and Photos, which enable the user to filter and choose properties that suit them. Customers start the reservation process with the help of the Booking entity and connect with the Payment and Reservation entities to make payments. There are check-in/out processing, recording of transactions and viewing of booking history. The Review table, which is related to the ReviewerID and ReviewedID of the User table, enables the guests to give feedback after staying, which increases the trust in the community and accountability of the platform.

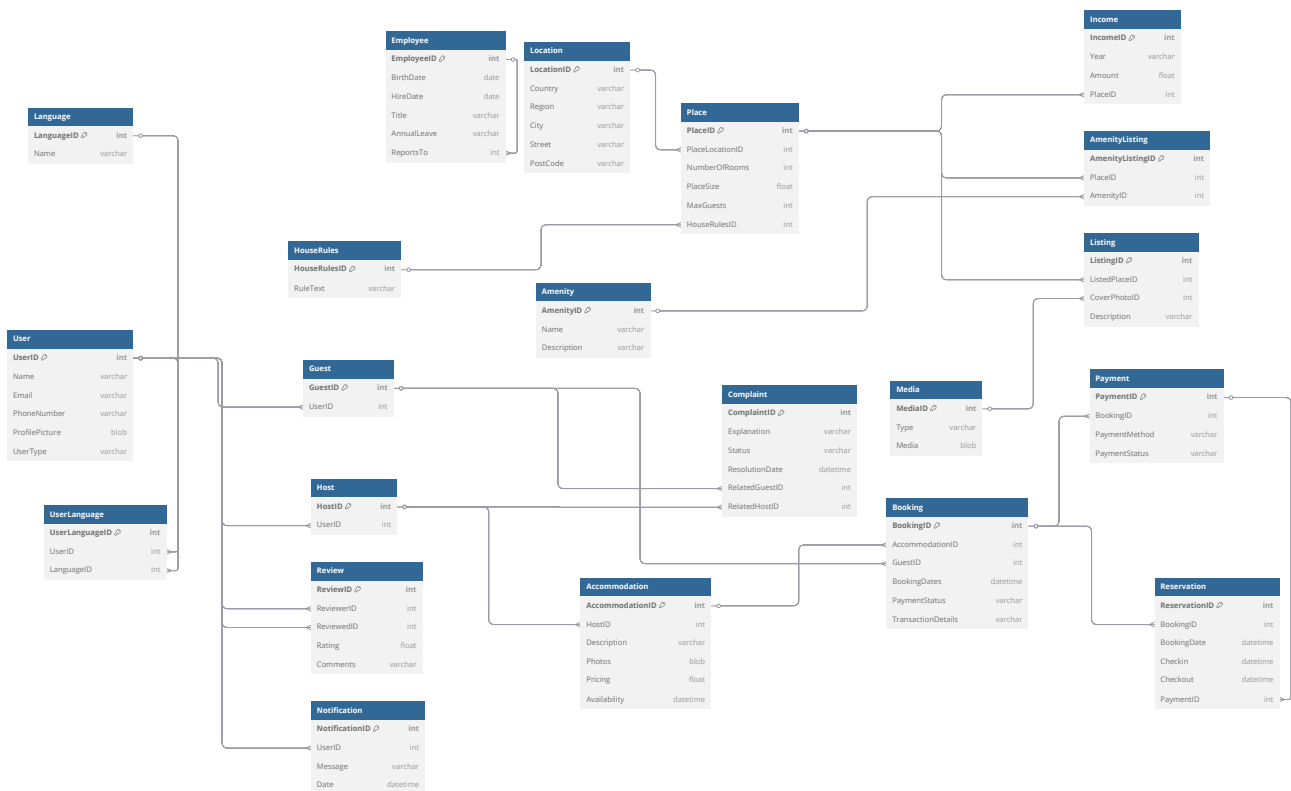
Another significant user group is hosts, who work with such functions as listing management, visualizing of earnings, and communicating with guests. The Listing, Place and AmenityListing entities give the host the capability of specifying the property attributes, related amenities and relevant HouseRules. Also, the Income table provides hosts with information about revenues in the long-term to assess the business performance. The mechanisms to resolve complaints are also implemented; the Complaint table links guests and hosts to dispute data and resolution tracking, increasing the platform transparency and user support.

The administrative and support staff functions of the platform are supported by the higher-level system-level entities like Employee, Notification and UserLanguage. The multilingual feature is done in a many-to-many relationship using the UserLanguage junction table and thus the platform can support internationalization. Event-driven communication is automated through notifications that are associated with UserID and assist in system alerts, reservation confirmations, and policies. The hierarchies of employees can be modeled through a self-referencing ReportsTo relationship, which allows role-based authorizations and delegation of tasks.

At the system level, data integrity and modularity is maintained by foreign key relationship between entities, Booking -> Accommodation, Payment -> Booking, and Reservation -> Payment. The management of the media is also centralized through the Media entity, which is being referenced by Listing through CoverPhotoID, so the media assets can be stored and retrieved efficiently. Such an extensive relational arrangement is consistent with academic research on the strong platform data design, which places a particular focus on the significance of normalized schemas regarding security, performance, and scalability (Ozdemir & Turker, 2019).

In this way, the data and the related functions do not only fulfill the expectations of the users in terms of their roles but also support platform governance, communication, and extensibility at the backend.

5. ERM (Entity Relationship Model):



Triple & Recursive Relationships:

The triple and recursive relations in the given ERM diagram are the key patterns of interactions in a solid accommodation platform.

The triple relationships are seen when there are three entities which are related to each other in a mutual context. As an example, in the BookingGuestProperty structure, there is an obvious triangular relationship between a Guest, the Booking they make and the Property (via Accommodation and Place) they book. This is a multi-entity dependency in the booking flow. The other triple structure can be seen in the ReviewBookingGuest relationship, where a Guest writes a Review based on a Booking, not only capturing feedback, but also putting it in context of the guest experience and booking event. All these interdependencies guarantee data consistency and deep analytics of user behavior and property performance.

On the scale of recursive relationships, the ERM discloses such intricacy in entities such as User, especially in support interactions. A User (Host or Guest) can engage with support in a cyclic way, by opening several support tickets and getting feedback or actions by support staff, which are also users. This User User Support Ticket User logic presents a self-referencing loop, and that is a feature of recursive relationships in which the entity is related to itself in more than one capacity. Likewise, recursive relationships can be found in the Employee table in the field called ReportsTo, indicating

that an employee (which can be a support or management entity) can report to an employee, creating a loop in the organization hierarchy.

The triple and recursive relationship mappings are sophisticated representations of relationships, which are carefully designed in order to represent subtle real-life interactions and decision paths, which increases the functional fidelity and analytical strength of the system.

Data Dictionary:

User Table

Attribute Name	Data Type	Description
UserID (PK)	int	A unique identifier assigned to each user.
Name	varchar	Full name of the user.
Email	varchar	Email address used for user communication.
PhoneNumber	varchar	Contact number of the user.
ProfilePicture	blob	Binary data representing user's profile photo.
UserType	varchar	Specifies whether the user is a host or guest.

Guest Table

Attribute Name	Data Type	Description
GuestID (PK)	int	Primary key uniquely identifying a guest.
UserID (FK)	int	Foreign key referring to the user table.

Host Table

Attribute Name	Data Type	Description
HostID (PK)	int	Unique identifier for a host profile.

Attribute Name	Data Type	Description
UserID (FK)	int	Links host to a specific user entry.

Employee Table

Attribute Name	Data Type	Description
EmployeeID (PK)	int	Unique key for an employee.
BirthDate	date	Date of birth of the employee.
HireDate	date	Date the employee officially joined.
Title	varchar	Designation or job title of the employee.
AnnualLeave	varchar	Information on allotted vacation days.
ReportsTo (FK)	int	Supervisor or manager reference in employee table.

Language Table

Attribute Name	Data Type	Description
LanguageID (PK)	int	Unique ID for each language record.
Name	varchar	Language name (e.g., English, Spanish).

UserLanguage

Attribute	Description	Data Type
UserLanguageID	Primary key to uniquely identify user-language relation.	int
UserID	Foreign key linking to a user entry.	int
LanguageID	Foreign key linking to a language entry.	int

Review

Attribute	Description	Data Type
ReviewID	Unique identifier for the review entry.	int
ReviewerID	References the user who posted the review.	int
ReviewedID	Refers to the user being reviewed.	int
Rating	Numeric score assigned in the review.	float
Comments	Written feedback or opinion included in the review.	varchar

Notification

Attribute	Description	Data Type
NotificationID	Identifier to distinguish each notification.	int
UserID	Foreign key connecting to the user who receives the notification.	int
Message	Text content delivered in the notification.	varchar
Date	Timestamp for when the notification was issued.	datetime

Accommodation

Attribute	Description	Data Type
AccommodationID	Unique identifier for the accommodation record.	int
HostID	Foreign key pointing to the host offering this accommodation.	int
Description	Brief overview of the accommodation features.	varchar
Photos	Image data or file representing accommodation visuals.	blob
Pricing	The cost per booking or night.	float
Availability	Dates and times the accommodation is available.	datetime

Booking

Attribute	Description	Data Type
BookingID	Primary key for each booking transaction.	int
AccommodationID	Foreign key linking to the booked accommodation.	int
GuestID	Foreign key referencing the user who made the booking.	int
BookingDates	Period of reservation including check-in and check-out.	datetime
PaymentStatus	Describes whether the booking payment is complete, pending, or failed.	varchar
TransactionDetails	Additional notes or data related to the transaction.	varchar

Payment

Attribute	Description	Data Type
PaymentID	Primary key uniquely identifying a payment.	int
BookingID	References the associated booking record.	int
PaymentMethod	Describes the method used for payment.	varchar
PaymentStatus	Current status of the payment transaction.	varchar

Reservation

Attribute	Description	Data Type
ReservationID	Unique identifier for each reservation.	int
BookingID	Foreign key pointing to the related booking.	int
BookingDate	Date the reservation was made.	datetime
Checkin	Scheduled check-in date and time.	datetime
Checkout	Scheduled check-out date and time.	datetime

Attribute	Description	Data Type
PaymentID	References the related payment record.	int

Complaint

Attribute	Description	Data Type
ComplaintID	Unique ID for each complaint entry.	int
Explanation	Details provided about the complaint.	varchar
Status	Current resolution status of the complaint.	varchar
ResolutionDate	Date when the complaint was addressed.	datetime
RelatedGuestID	Guest ID involved in the complaint.	int
RelatedHostID	Host ID involved in the complaint.	int

Location

Attribute	Description	Data Type
LocationID	Primary key for each location.	int
Country	Country of the location.	varchar
Region	State or regional subdivision.	varchar
City	City name.	varchar
Street	Street address.	varchar
PostCode	Postal or ZIP code.	varchar

Place

Attribute	Description	Data Type
PlaceID	Unique identifier for a place entity.	int
PlaceLocationID	Foreign key to the location of the place.	int
NumberOfRooms	Number of rooms available in the place.	int
PlaceSize	Measurement of the place's area.	float
MaxGuests	Maximum number of guests allowed.	int
HouseRulesID	Foreign key linking to applicable house rules.	int

Listing

Field Name	Data Type	Description
ListingID	int	Primary key, uniquely identifies a listing
ListedPlaceID	int	Foreign key referencing the listed place
CoverPhotoID	int	Foreign key referencing media used
Description	varchar	Textual details about the listing

Media

Field Name	Data Type	Description
MediaID	int	Primary key for media assets
Type	varchar	Type of media (image, video, etc.)
Media	blob	Binary large object storing the file

HouseRules

Field Name	Data Type	Description
HouseRulesID	int	Primary key, uniquely identifies rules
RuleText	varchar	Text of the house rule

Income

Field Name	Data Type	Description
IncomeID	int	Primary key, uniquely identifies income
Year	varchar	Year in which income was recorded
Amount	float	Total income amount
PlaceID	int	Foreign key referencing related property

Amenity

Field Name	Data Type	Description
AmenityID	int	Primary key for amenities
Name	varchar	Name or title of the amenity
Description	varchar	Detailed description of the amenity

AmenityListing

Field Name	Data Type	Description
AmenityListingID	int	Primary key for amenity-place association
PlaceID	int	Foreign key referencing the place entity
AmenityID	int	Foreign key referencing the amenity entity

5. Summary

The thorough ERM (Entity Relationship Model) created of Airbnb is a carefully thought-out and organized data model that serves the complex nature of a massive online accommodation service. The design includes a broad scope of entities and each entity is a representation of the main components of operation, including users (guests and hosts), employees, locations, accommodations, reservations, and transactions. Such architecture allows accurate data flow capture and management throughout the system. The interconnectedness between the entities, i.e., the connections between bookings and payments, guests and reviews, hosts and accommodations, and places and amenities, demonstrates the extent of normalization and integrity imposed on the model.

In addition, the use of associative tables such as UserLanguage and AmenityListing also makes many-to-many relationships to be managed effectively, enhancing flexibility and scalability. User interaction and transparency of operations is facilitated by logical extensions of Review, Complaint and Notification modules. The application of the Chen notation assisted in establishing cardinalities and recursion in the required areas particularly in situations that involved Employee hierarchy or Complaint associations. On the whole, this ERM does not only facilitate the scope of the Airbnb operations, but it also guarantees flexibility to the future changes, consistency and clarity of the data within a complex transactional environment.

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